United International **University**

B.Sc. in Computer Science & Engineering (CSE)

Course Syllabus

<table>
<tr><td>1</td><td>Course Title</td><td>Data Communication</td></tr>
<tr><td>2</td><td>Course Code</td><td>CSE 3715</td></tr>
<tr><td>3</td><td>Trimester</td><td>Summer 2025</td></tr>
<tr><td>4</td><td>Pre-requisites</td><td>-</td></tr>
<tr><td>5</td><td>Credit Hours</td><td>3.00</td></tr>
<tr><td>6</td><td>Class Information</td><td>
<table>
<tr><th>Section</th><th>Day</th><th>Time</th><th>Room</th></tr>
<tr><td>B</td><td>Sunday, Wednesday</td><td>12.30 PM - 01.50 PM</td><td>329</td></tr>
</table>
</td></tr>
<tr><td>7</td><td>Instructor Information</td><td>
<table>
<tr><th>Name</th><td>Abdullah Ibne Masud Mahi</td></tr>
<tr><th>Code</th><td>AIMM</td></tr>
<tr><th>Email</th><td>ibnemasud@cse.uiu.ac.bd</td></tr>
<tr><th>Phone</th><td>+88 01842 93 91 94</td></tr>
<tr><th>Office</th><td>919</td></tr>
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<tr><td>8</td><td>Counselling Hours</td><td>
<table>
<tr><th>Day</th><th>Time</th></tr>
<tr><td>Saturday</td><td>11.10 AM - 12.30 PM, 03.10 PM - 04.30 PM</td></tr>
<tr><td>Sunday</td><td>11.10 AM - 12.30 PM</td></tr>
<tr><td>Monday</td><td>11.10 AM - 04.30 PM</td></tr>
<tr><td>Tuesday</td><td>11.10 AM - 12.30 PM, 03.10 PM - 04.30 PM</td></tr>
<tr><td>Wednesday</td><td>11.10 AM - 12.30 PM</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Text Book</td><td>Data Communication and Networking (4th Edition) - Behrouz A. Forouzan</td></tr>
<tr><td>10</td><td>References</td><td>Computer Networks: A Top Down Approach(5th Edition) - Jim Kurose</td></tr>
</table>

11	**Teaching Methods**	Lecture and case studies.

Course Outline

<table>
<tr><td>12</td><td>Course Contents (UGC Approved)</td><td>Introduction to Data Communication; Analog and digital signals and their role in data transmission; OSI and TCP/Layered Model; Transmission media and Channel Capacity; Line Coding Techniques; Modulation and Multiplexing of signals for data transmission; Error detection and Error correction techniques; Cellular Technology; Wireless Communication.</td></tr>
<tr><td>13</td><td>Course Outcomes (COs)</td><td>
<table>
<tr><th>COs</th><th>Description</th></tr>
<tr><td>CO1</td><td>Learn fundamentals of data communication. Explain, describe and design layered architecture of communication according to OSI and TCP/IP model.</td></tr>
<tr><td>CO2</td><td>Measure information content in a signal and convert a time domain signal to frequency domain version.</td></tr>
<tr><td>CO3</td><td>Analyze various Encoding Techniques. Describe different modulation and multiplexing techniques to send signals in different forms using different properties.</td></tr>
<tr><td>CO4</td><td>Learn error detection and error correction techniques.</td></tr>
<tr><td>CO5</td><td>Describe the basic principles of Cellular Technology and Wireless Communication.</td></tr>
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<tr><td>14</td><td>CO with Assessment Methods</td><td>
<table>
<tr><th>CO</th><th>Assessment Method</th><th>Percentage (%)</th></tr>
<tr><td>-</td><td>Attendance</td><td>5</td></tr>
<tr><td>-</td><td>Assignments</td><td>5</td></tr>
<tr><td>-</td><td>Class Tests</td><td>20</td></tr>
<tr><td>CO1, CO2, CO3</td><td>Midterm Exam</td><td>30</td></tr>
<tr><td>CO3, CO4, CO5</td><td>Final Exam</td><td>40</td></tr>
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15	**Mapping of COs and Program Outcomes**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	C											
CO2		C										
CO3		C										
CO4	C											
CO5				C								

Class Outline

Class	Topics	COs	Outcome
1	Introduction to Data Communication	1	Define data communication and its components, identify real-world applications
2, 3	Layered Network Architecture (OSI, TCP/IP)	1	Compare OSI and TCP/IP models, describe the role of each layer
4	Analog & Digital Signals	2	Differentiate analog and digital signals, explain amplitude, frequency, and phase
5	Transmission Impairments	2	Identify several signal impairments (attenuation, distortion, noise)
6	Data Rate Limits	2	Apply Nyquist and Shannon formulas to compute maximum data rates
7, 8, 9	Digital Transmission	3	Describe and compare line coding schemes, analyze their properties, Explain PCM & Sampling
10	Analog Transmission	3	Explain ASK, FSK, PSK, QAM, AM, PM, FM
11	Transmission Media	2	Compare twisted pair, coax, fiber and wireless media
12	Mid Exam Review		
MID-TERM EXAM			
13, 14	Multiplexing Techniques	3	Compare FDM, TDM, WDM, explain their applications
15, 16	Switching Techniques	3	Compare circuit, packet, and message switching, explain their applications
17, 18	Error Detection Techniques	4	Apply CRC, Parity Check, and Checksums for error detection
19	Error Correction Techniques	4	Apply Hamming Code for error correction
20, 21	Multiple Access Techniques	3	Learn several multiple access (ALOHA, CSMA/CD, CSMA/CA) techniques
22, 23	Cellular Concept	5	Learn cellular system design fundamentals (Cell, Frequency Reuse, Handoff)
24	Final Exam Review		
FINAL EXAM			

Appendix

Grading Policy

Letter Grade	Marks	Grade Point	Letter Grade	Marks	Grade Point
A (Plain)	90-100	4.00	C- (Minus)	62-65	1.67
A- (Minus)	86-89	3.67	D+ (Plus)	58-61	1.33
B+ (Plus)	82-85	3.33	D (Plain)	55-57	1.00
B (Plain)	78-81	3.00	F (Fail)	<55	0.00
B- (Minus)	74-77	2.67	I	-	0.00
C+ (Plus)	70-73	2.33	W	-	0.00
C (Plain)	66-69	2.00	R	-	0.00

Program Outcomes

Program Outcomes	
1	**Engineering knowledge:** Apply knowledge of mathematics, natural science, engineering fundamentals, and Computer Science and Engineering to the solution of complex engineering problems.
2	**Problem analysis:** Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3	**Design/development of solutions:** Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4	**Investigation:** Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
5	**Modern tool usage:** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
6	**The engineer and society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems.
7	**Environment and sustainability:** Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts.
8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
9	**Individual work and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**Project management and finance:** Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's 13 own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.